

Chapter 1e: Compactness & Connectedness

Pathak - An Introduction to Nonlinear Analysis and Fixed Point Theory

Lecture Notes

Section 1.5: Compactness (pp. 59–68)

July 14, 2026

Chapter Overview

- 1 Fundamentals of Compactness
- 2 Key Theorems on Compact Sets
- 3 Weak Compactness
- 4 Compact Operators
- 5 Properties of Compact Sets
- 6 Applications and Examples
- 7 Numerical Examples
- 8 Summary and Key Takeaways

What is Compactness?

Intuitive Approach

Main Idea

Compactness generalizes the properties of closed and bounded sets in finite-dimensional spaces to arbitrary topological spaces.

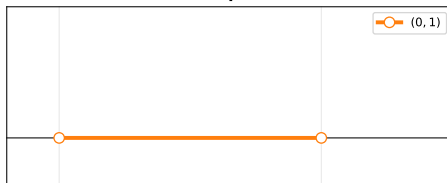
Key Properties in Compact Sets:

- Every sequence contains a convergent subsequence (sequential compactness)
- Closed and bounded in complete metric spaces (Heine-Borel)
- Continuous images of compact sets are compact
- Closed subsets of compact sets are compact

(a) Compact Set in $\mathbb{R}$



(b) Non-compact Set in $\mathbb{R}$



Sequential Compactness

Definition 1.29 & Key Characterization

Definition (Sequentially Compact)

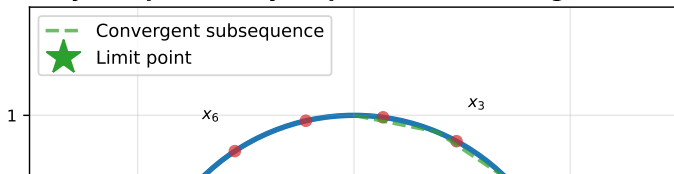
A topological space in which every sequence has a convergent subsequence is called **sequentially compact**.

Proposition (1.16)

Sequential compactness is equivalent to compactness in the case of a **metric space**.

Visualization:

Sequentially Compact: Every Sequence has Convergent Subsequence



The Heine-Borel Theorem

Characterization in Euclidean Spaces

Theorem (1.25 - Heine-Borel)

A subset C of $\mathbb{R}$ is compact if and only if it is **closed and bounded**.

Corollary (1.5)

A subset C of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Examples:

- $C = [0, 1] \subset \mathbb{R}$ is **compact** (closed and bounded)
- $(0, 1) \subset \mathbb{R}$ is **not compact** (open, not closed)
- $\mathbb{R}$ is **not compact** (unbounded)
- $\mathbb{R}^n$ for $n \geq 1$ is **not compact** (unbounded)

Continuous Images Preserve Compactness

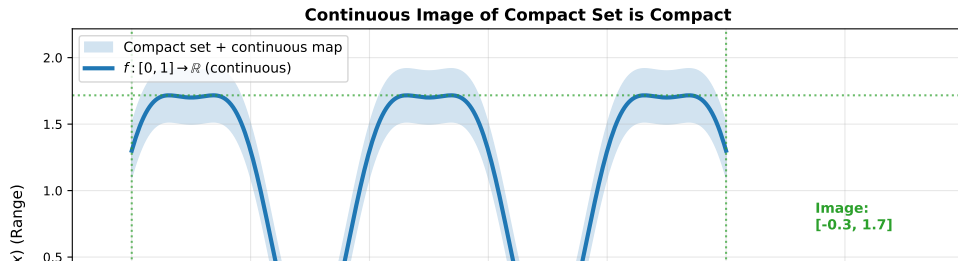
Theorem 1.24

Theorem (1.24)

The continuous image of a compact metric space is compact.

Consequence: If X is compact and $f : X \rightarrow Y$ is continuous, then:

- $f(X)$ is compact
- f attains its maximum and minimum on X
- f is uniformly continuous on X (Theorem 1.26)



Uniform Continuity on Compact Spaces

Theorem 1.26

Theorem (1.26)

Let T be a continuous mapping from a compact metric space (X, d) into another metric space (Y, ρ) . Then T is **uniformly continuous**.

Interpretation:

$$\forall \epsilon > 0, \exists \delta > 0 : d(x, x') < \delta \implies \rho(T(x), T(x')) < \epsilon$$

The δ depends only on ϵ and not on the point x .

Applications:

- Error estimates in numerical analysis
- Convergence of approximation schemes
- Perturbation theory

Relatively Compact and Locally Compact

Definition 1.30

Definition (1.30)

A subset C of a topological space is called:

- **Relatively compact** if its closure $\bar{C}$ is compact
- **Locally compact** if each point has a compact neighborhood

Example:

- $(0, 1)$ is relatively compact in $\mathbb{R}$ since $\overline{(0, 1)} = [0, 1]$ is compact
- $\mathbb{R}$ is locally compact (every point has compact neighborhood)
- ℓ^2 is not locally compact

Definition (1.30)

A topological space is locally compact if each point has a compact neighborhood.

Mazur's Theorem

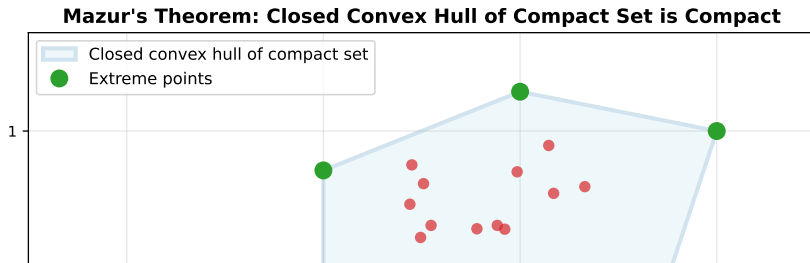
Theorem 1.27

Theorem (1.27 - Mazur)

The closed convex hull $\overline{\text{co}}(C)$ of a compact set C of a Banach space is compact.

Implications:

- Convex combinations of points in compact sets stay in compact set
- The convex hull operation preserves compactness
- Useful in variational methods



Finite Dimensionality Characterization

Proposition 1.17

Proposition (1.17)

A normed space X is finite dimensional if and only if every closed and bounded subset of X is compact.

Key Insight:

- In finite dimensions: closed + bounded $\Rightarrow$ compact
- In infinite dimensions: this equivalence **fails**
- Example: $B_X = \{x \in \ell^2 : \|x\|_2 \leq 1\}$ is closed and bounded but NOT compact

Important Observation

The failure of Heine-Borel in infinite dimensions is a fundamental distinction that motivates the study of weak compactness and other generalized notions of compactness.

Weak Compactness in Banach Spaces

Theorems 1.28 – 1.31

Theorem (1.28 - Eberlein-Smulian)

A weakly closed subset C of a Banach space X is weakly compact if and only if C is weakly sequentially compact.

Theorem (1.29 - Kakutani)

A Banach space X is reflexive if and only if the closed unit ball $\mathbb{B}_X = \{x \in X : \|x\| \leq 1\}$ is weakly compact.

Theorem (1.30)

A Banach space X is reflexive if and only if every closed convex bounded subset of X is weakly compact.

Weak Compactness: Weaker Topology $\Rightarrow$ Larger Compact Sets

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Strong topology ball

Weak Compactness vs Strong Compactness

Theorem 1.31

Theorem (1.31)

Let X be a reflexive Banach space and C a subset of X . Then C is weakly compact (compactness in weak topology) if and only if C is bounded (boundedness in strong topology).

Key Difference:

- **Strong compactness:** Very restrictive, rare in infinite dimensions
- **Weak compactness:** More abundant due to Kakutani's theorem

For Reflexive Banach Spaces:

$$\text{Bounded} \iff \text{Weakly Compact}$$

Remark

Theorem 1.33 (Banach-Alaoglu): For any normed space X , the unit ball $\mathbb{B}_{X^*}$ in the dual space X^* is compact in the weak* topology.

Characterization of Reflexivity

Theorem 1.32

Theorem (1.32)

For a Banach space X , the following are equivalent:

- (a) X is reflexive
- (b) $\mathbb{B}_X$ is weakly compact
- (c) Every bounded sequence in X (strong topology) has a weakly convergent subsequence
- (d) X^* is reflexive
- (e) For any $f \in X^*$, there exists $x \in \mathbb{B}_X$ such that $f(x) = \|f\|_*$
- (f) If $\{C_n\}$ is a descending sequence of nonempty closed convex subsets, then $\bigcap_{n=1}^{\infty} C_n \neq \emptyset$
- (g) $\sigma(X^*, X) = \sigma(X^*, X^{**})$ (weak and weak* topologies coincide)

Examples of Reflexive Spaces:

- $L^p[0, 1]$ for $1 < p < \infty$
- ℓ^p for $1 < p < \infty$
- Sobolev spaces $W^{k,p}(\Omega)$ for $1 < p < \infty$

Compact Operators

Definition and Properties

Definition (Compact Operator)

Let X and Y be normed spaces. A bounded linear operator $A : X \rightarrow Y$ is called **compact** if the image of the closed unit ball $\mathbb{B}_X$ is relatively compact in Y , i.e., $\overline{A(\mathbb{B}_X)}$ is compact.

Equivalent Characterization: A is compact $\iff$ every bounded sequence in X has a subsequence whose image under A converges in Y .

Key Properties:

- Compact operators map bounded sets to relatively compact sets
- Products of compact operators with bounded operators are compact
- Limits of sequences of compact operators are compact

Theorem 1.37

If $F_n : X \rightarrow Y$ is a sequence of compact operators such that $\lim_{n \rightarrow \infty} \|F_n x - F x\| = 0$ uniformly for x in every bounded subset of X , then F is also compact.

Compact Operators in L^2 Spaces

Example 1.37

Example: The integral operator

$$[Fx](s) = \int_0^{2\pi} x^2(t) dt, \quad s \in [0, 2\pi]$$

on $X = L_2[0, 2\pi]$ is continuous and compact, but not completely continuous.

Proof Sketch:

① For the ball $\mathbb{B}_r = \{x \in L_2[0, 2\pi] : \|x\| \leq r\}$:

$$|Fx(s)| = \left| \int_0^{2\pi} x^2(t) dt \right| \leq \int_0^{2\pi} r^2 dt = 2\pi r^2$$

② Functions in $F(\mathbb{B}_r)$ are uniformly bounded

③ By Ascoli-Arzelà theorem: $F(\mathbb{B}_r)$ has convergent subsequence in $L_2[0, 2\pi]$

④ Therefore F is compact

Completely Continuous Operators

Definition 1.39 & Theorem 1.38

Definition (1.39)

An operator $F : X \rightarrow Y$ is called **completely continuous** if for any sequence $\{x_n\}$ converging weakly to x , the sequence $\{Fx_n\}$ converges to Fx .

That is: $x_n \rightharpoonup x \implies Fx_n \rightarrow Fx$.

Theorem (1.38)

If F is a completely continuous operator from a reflexive Banach space X into a Banach space Y , then F is both continuous and compact.

Corollary

For **linear operators** from a reflexive Banach space to another Banach space:

$$\text{Completely continuous} \iff \text{Compact}$$

(provided the domain is reflexive)

Degenerate Integral Operators

Compact Linear Operators

Definition (Degenerate Integral Operator)

An integral operator $K : C([0, 1]) \rightarrow C([0, 1])$ defined by

$$[Kf](x) = \int_0^1 k(x, y) f(y) dy$$

is called **degenerate** if $k(x, y)$ is a finite sum of separated terms:

$$k(x, y) = \sum_{i=1}^n \varphi_i(x) \psi_i(y)$$

where $\varphi_i, \psi_i : [0, 1] \rightarrow \mathbb{R}$ are continuous and linearly independent.

Hilbert-Schmidt Kernel: For a domain $\Omega \subset \mathbb{R}^n$, a kernel $k : \Omega \times \Omega \rightarrow \mathbb{C}$ is a **Hilbert-Schmidt kernel** if

$$\int_{\Omega} \int_{\Omega} |k(x, y)|^2 dx dy < \infty$$

General Properties of Compact Sets

Summary of Key Results

In a Normed Space X

Every compact subset $C \subseteq X$ has the following properties:

- 1 C is **closed**
- 2 C is **bounded**
- 3 C is **complete**
- 4 C is **separable** (contains a countable dense subset)

Important Remark:

- The **converse is NOT true** in infinite dimensions
- Closed + bounded $\nRightarrow$ compact in infinite-dimensional normed spaces
- This is the fundamental difference from finite-dimensional Euclidean spaces

Properties of Compact Sets

Closed Sets and Compact Sets

Key Relationships

Fact 1

Every compact subset of X is closed, but the converse need not be true.

Example: ℓ^2 is closed in itself but not compact.

Fact 2

A closed subset of a compact set is compact.

Fact 3

If C is a nonempty subset of a Banach space X and $\{C_n\}$ is a descending sequence of nonempty closed convex subsets with $C_n \subseteq C$ for all n , then:

$$\bigcap_{n=1}^{\infty} C_n \neq \emptyset \quad \text{iff } X \text{ is reflexive}$$

Compactness in Differential Equations

Application Example

Existence Problem: Consider the integral equation

$$u(t) = \int_0^t K(t,s)f(u(s)) ds, \quad t \in [0, T]$$

Solution Strategy:

- ① Rewrite as fixed point problem: $u = Tu$ where T is a suitable operator
- ② Show that T maps a convex compact set to itself
- ③ Apply Schauder fixed point theorem (requires T to be continuous)
- ④ Compactness is often obtained via:
 - Ascoli-Arzelà theorem (for function spaces)
 - Properties of integral operators
 - Weak compactness in reflexive spaces

Key Role of Compactness

Compactness provides a weak form of finiteness that allows infinite-dimensional fixed point theorems to work analogously to Brouwer's theorem in $\mathbb{R}^n$.

Compactness in Optimization

Existence of Extrema

Extremum Problem: Minimize a continuous functional $J : X \rightarrow \mathbb{R}$ over a set $C \subseteq X$.

Standard Approach:

- 1 If C is compact and J is continuous, then:

$$\exists x_0 \in C : J(x_0) = \min_{x \in C} J(x)$$

- 2 If C is closed, convex, and bounded in a **reflexive** Banach space:
 - C is weakly compact
 - If J is weakly lower semicontinuous, minimum exists

Why Compactness Matters:

- Without compactness: infimum may not be attained
- Compactness + continuity = guaranteed minimum
- Weak compactness + weak lower semicontinuity = minimum in infinite dimensions

Compactness in Control Theory

Bang-Bang Control Problems

Control System:

$$\dot{x}(t) = f(t, x(t), u(t)), \quad x(0) = x_0, \quad u(t) \in U$$

where U is a compact control set.

Key Results:

- ① If U is compact and f is continuous and locally Lipschitz in x :
 - Solution trajectories form a compact set
 - Optimal control exists
- ② The set of reachable points from initial condition is compact
- ③ Value function of optimal control problem is continuous

Application

Optimal control problems in bounded domains with compact control constraints are guaranteed to have optimal solutions.

Numerical Example: Compactness Test in $\mathbb{R}^n$

Python Implementation

```
import numpy as np

# Example: Check if sets are compact
def is_compact_euclidean(A, is_closed, is_bounded):
    """In  $\mathbb{R}^n$ , compact iff closed and bounded"""
    return is_closed and is_bounded

# Test cases
sets = {
    '[0,1]': (True, True),           # Compact
    '(0,1)': (False, True),          # Not compact (open)
    '[0,inf)': (True, False),        # Not compact (unbounded)
    'Open_disk': (False, True),      # Not compact
}

print("Compactness in  $\mathbb{R}^n$  (Heine-Borel Theorem):")
for name, (closed, bounded) in sets.items():
    compact = is_compact_euclidean(name, closed, bounded)
    print(f"{name:15} Closed: {closed}, Bounded: {bounded} => "
          f"Compact: {compact}")

# Unit ball in  $l^2$  (infinite dimensions)
print("\nIn infinite dimensions ( $l^2$  space):")
print("Closed unit ball: Closed: True, Bounded: True")
print("Compact in strong topology: FALSE")
print("Compact in weak topology: TRUE (Kakutani)")
```

Numerical Example: Continuous Functions on Compact Sets

Properties and Computation

```
import numpy as np
from scipy.optimize import minimize_scalar

# Continuous function on compact domain [0,1]
def f(x):
    return np.sin(5 * np.pi * x) + 0.3 * np.cos(10 * np.pi * x)

# Properties guaranteed by Theorem 1.24:
# 1. f attains its maximum and minimum on [0,1]
# 2. f([0,1]) is a compact interval [m, M]

# Find extrema
x_test = np.linspace(0, 1, 1000)
y_test = f(x_test)

min_val = np.min(y_test)
max_val = np.max(y_test)
x_min = x_test[np.argmin(y_test)]
x_max = x_test[np.argmax(y_test)]

print(f"Domain: [0,1] (compact interval)")
print(f"Function: f(x) = sin(5 x) + 0.3*cos(10 x)")
print(f"\nMinimum: f({x_min:.3f}) = {min_val:.4f}")
print(f"Maximum: f({x_max:.3f}) = {max_val:.4f}")
print(f"Image f([0,1]) = [{min_val:.4f}, {max_val:.4f}] (compact)")
```


Numerical Example: Weak Compactness in ℓ^2

Bounded vs Compact

```
import numpy as np

# In  $\ell^2$ , reflexive Banach space
# Kakutani: Bounded iff Weakly Compact

# Generate bounded sequence
np.random.seed(42)
T = 100 # number of time points
N = 50  # dimension of approximation

# Sequence in unit ball of  $\ell^2$ 
sequences = []
for n in range(20):
    seq = np.random.randn(N) / np.sqrt(N)
    norm = np.linalg.norm(seq)
    sequences.append(seq / norm) # normalize to unit ball

sequences = np.array(sequences)

print("Bounded sequence in  $\ell^2$  unit ball:")
print(f"Number of sequences: {len(sequences)}")
print(f"Norm of each: {[np.linalg.norm(s) for s in sequences[:5]]}")

print("\nProperties:")
print("- Strong topology: NOT compact")
print("- (but has weakly convergent subsequence)")
```

Key Theorems Summary

Section 1.5 - Compactness

Fundamental Results

- 1 **Heine-Borel (Thm 1.25):** In $\mathbb{R}^n$: compact $\iff$ closed + bounded
- 2 **Sequential Compactness (Prop 1.16):** In metric spaces: sequential compactness $\iff$ compactness
- 3 **Continuous Image (Thm 1.24):** Continuous image of compact set is compact
- 4 **Uniform Continuity (Thm 1.26):** Continuous map on compact is uniformly continuous
- 5 **Mazur (Thm 1.27):** Closed convex hull of compact set is compact

Weak Compactness (Infinite Dimensions)

- 1 **Kakutani (Thm 1.29):** Reflexive $\iff$ closed unit ball weakly compact
- 2 **Eberlein-Smulian (Thm 1.28):** Weak sequential compactness $\iff$ weak compactness
- 3 **Banach-Alaoglu (Thm 1.33):** Unit ball in dual is weak* compact

Compact vs. Non-Compact Characterization

Quick Reference

Set Property	Compact	Non-Compact
$[0,1]$	✓	
$(0,1)$		✓
Closed unit ball in $\mathbb{R}^n$	✓	
Closed unit ball in ℓ^2 (strong)		✓
Closed unit ball in ℓ^2 (weak)	✓	

Decision Tree

- Is set closed and bounded in $\mathbb{R}^n$? $\rightarrow$ YES = Compact
- Is domain a complete metric space? $\rightarrow$ Apply Heine-Borel
- Working in infinite-dimensional Banach space? $\rightarrow$ Check weak topology
- Is space reflexive? $\rightarrow$ Bounded sets are weakly compact

Why Compactness Matters

Applications in Analysis

In Functional Analysis

- **Fixed Point Theorems:** Schauder fixed point theorem requires compact operators
- **Existence Theorems:** Guarantees solutions to differential equations
- **Optimization:** Ensures extrema of continuous functionals are attained

In Topology

- Compactness is a topological property (preserved under continuous maps)
- Local compactness is important for partition of unity
- Paracompactness relates to existence of good coverings

Bridge Between Finite and Infinite Dimensions

- Provides "infinite-dimensional analogue" of closed and bounded
- Weak compactness enables existence results without strong assumptions
- Critical for modern nonlinear analysis

References and Further Reading I

From Pathak's Book

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Thank You

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Questions and Discussion Welcome